



## Comparison of manual vs. speech-based interaction with in-vehicle information systems

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### ABSTRACT

This study examined whether speech-based interfaces for different in-vehicle-information-systems (IVIS) reduce the distraction caused by these systems. For three frequently used systems (audio, telephone with name selection, navigation system with address entry and point of interest selection) speech, manual control and driving without IVIS (baseline) were compared. The Lane Change Task was used to assess driving performance. Additionally, gaze behavior and a subjective measure of distraction were analyzed. Speech interfaces improved driving performance, gaze behavior and subjective distraction for all systems with the exception of point-of-interest entry. However, these improvements were overall not strong enough to reach the baseline performance level. Only in easy segments of the driving task the performance level was comparable to baseline. Thus, speech-based IVIS have to be further developed to keep the cognitive complexity at an adequate level which does not disturb driving. However, looking at the benefits, speech control is a must for the car of the future.

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### 1. Introduction

Distraction of the driver is one of the major causes for car accidents. The National Highway Traffic Safety Administration (Wang et al., 1996; Ranney et al., 2000) estimates that approximately 25% of the accidents in the years 1995–1999 which were registered by the police were due to inattention of the driver. Further analysis identified some specific sources of distraction which were “adjusting the radio, cassette or CD” comprising 11.4% of the crashes, “other device/object in the vehicle” with 2.9% and “using a cell phone” (e.g. listening, talking and dialing) with 1.5% (Stutts et al., 2001). These results were based on self-reports of drivers with accidents. Thus, one may assume that these numbers underestimate the real scope of the problem. This assumption is supported by the real driving data analysis of Klauer et al. (2006) using the “100 car naturalistic driving study”. It was found that in 40% of all drives secondary tasks were conducted. Additionally, more and more systems and functions have become available in the car, e.g. playing CDs and DVDs, phoning, entering a destination in the navigation system, etc. This rapid development of increasingly sophisticated information systems is associated with more complex menu structures which add to the mental workload of the driver and which will probably further increase distraction.

The effects of engaging in these kinds of secondary tasks have been demonstrated in two simulator studies. Chisholm et al. (2008) found a significant increase of reaction time to several hazards (pedestrian walking into the path of car; parked vehicle pulling out of the side of the roadway; leading car braking suddenly) while selecting a specific song in an MP3 player. Furthermore, the number of glances and the glance duration away from the road were increased in a difficult MP3 player condition. Reed and Robbins (2008) conducted a simulator study to investigate the impact of text messaging while driving. Results show that participants' driving behavior was impaired, particularly reaction time (35% slower when writing a text message) and the ability to maintain lateral vehicle control. While texting, participants slowed down significantly, indicating that they were aware of the distraction and potential risk and tried to compensate for it. Apparently, these compensatory efforts were not successful.

These effects may be explained by the fact that humans have only a limited amount of available cognitive resources at any given time. However, according to the multiple resource theory of attention (Wickens, 2002) there are different pools of resources which can be tapped in parallel. These pools concern on the one hand input modalities (especially visual and acoustic input) and on the other hand reactions (manual vs. verbal reaction). Tasks which require the same resources will interfere. For example, operating a device via touch screen will interfere with driving a vehicle. From multiple resource theory, speech-based human–machine-interaction should be much better suited while driving because this taps different resources than driving.

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However, a large number of studies have demonstrated that speaking and listening may also impair driving (for an overview, see Horrey and Wickens, 2006). For example, Lee et al. (2001) investigated driving performance while operating a speech-based e-mail system. Results show a 30% increase in reaction time to a braking lead vehicle when the speech-based task was carried out. Furthermore, subjective workload increased significantly while performing the e-mail task compared to a baseline condition. A follow-up study of Jameson et al. (2004) confirmed these results and demonstrated a significant reduction of the time to collision in the distraction condition.

These findings do not only apply to human–machine-interaction by ways of speech, but also for human communication. In a meta-analysis of the effects of cell phone use on driving performance, Caird et al. (2008) included 33 studies through 2007. Results show a decrease of reaction time to events and stimuli while talking, independent of the research setting (laboratory, driving simulation or on-road driving). Furthermore, there were no differences for using handheld or hands-free speaking systems. The mean increase in reaction time was found to be 0.25 s for all types of phone-related tasks. No significant difference was found between different communication modes (talking to a fellow passenger vs. talking via phone) or types (naturalistic vs. information-processing speaking tasks).

Although human communication and speech interfaces can have a negative impact on driving safety, speech interfaces may still be better than displays and manual controls. Engaging in a conversation and operating a device via speech apparently use resources which are also required for driving. This interference might be minimized by reducing the requirements for visual input and manual control exerted by the secondary task. In order to examine this idea, a direct comparison of the effect of device control via speech and via display and manual control during car driving is required. This is the aim of the study presented here.

Using a simple driving simulation (the Lane Change Task; Mattes, 2003) four commercially available in-vehicle information systems (two different navigation devices, a hands-free mobile phone car kit, a music system) were compared operating these either by a touch screen or a speech interface. To give a meaningful estimation of the effect size both conditions were compared to a baseline. The two navigation systems differed in the way the speech interface was designed enabling us to also obtain some results on the importance of interface design to driving safety.

## 2. Method

### 2.1. Driving task

This study used the Lane Change Task (LCT, Mattes, 2003) to evaluate the distraction caused by the different in-vehicle information systems (IVIS). A standard PC equipped with a joystick steering-wheel, gas and brake pedal was used to conduct the experiment. In the LCT simulation, the driver sees a straight section of a three-lane road and is instructed to keep the current lane while driving at a constant speed of 60 km/h (the driver cannot drive faster than this). At some points signs are introduced which become legible at a certain distance. These indicate that the driver should change the lane (the target lane is shown) as soon as possible. One trial consists of 18 lane changes in a random order (left vs. right, movement across one lane vs. movement across two lanes) and takes about 3 min. For the analysis, the task was divided into separate lane-keeping and lane-changing phases. Driver behavior and car reactions were recorded with 62 Hz corresponding to a precision of 16 ms. For the lane-keeping phases, the standard deviation of the lateral position (SDLP) of the car was computed. For the lane changes, a reaction

time was measured from the point where the sign became legible to the point where the driver started to steer. Additionally, for the whole trial a mean deviation to an “ideal” behavior was computed. For this simplified ideal behavior it was assumed that during lane-keeping the car should stay in the middle of the lane. The lane change was assumed to begin 30 m before each sign and to take 10 m. A simple linear trajectory was used to change the lane. While this is not a typical lane change behavior of a human driver, this gives an easy-to-compute standard to which the real behavior can be compared. This computation of the mean deviation was adopted from Mattes (2003).

The idea in creating this task was to assess the performance in two basic components of the driving task, lane-keeping and changing the lane. Eighteen lane changes within 3 min amount to about one lane change every 10 s. If a driver was required to change the lane every 10 s in a real driving situation he or she would probably not engage in secondary tasks during that period of driving. From this point of view, the LCT is an artificial situation. However, the advantage of this task is to create a well-defined level of the driving-task difficulty which requires frequent attention of the driver. But this does not hinder the drivers to conduct secondary tasks at the same time. Of course, secondary task performance is not as good as in single task conditions, but this is also typical for natural behavior in the car. Thus, the LCT enables to compare different secondary task at a constant level of driving-task difficulty.

Besides the three driving parameters described above (SDLP, reaction time, mean deviation), gaze behavior and subjective distraction was measured after each condition. Drivers rated their perceived distraction in a two-step procedure. First, they selected of five verbal categories ranging from ‘very little’ to ‘very strong’. For the category selected they refined their assessment using a 3-point scale (–/0/+) (Heller, 1982). This procedure is very easy to do and gives reliable data on a 15-point-scale ( $5 \times 3$ ).

During the trials all participants were videotaped. Following the experiment, a video-rating of the gaze behavior was conducted using the software INTERACT. Gaze was classified as either ‘looking at the screen of the LCT’ or ‘looking away from the screen towards the secondary task’. From these data, the percentage of gaze duration away from the LCT (‘%gaze away’ as a percentage of the total driving time) was computed.

### 2.2. Secondary tasks

To compare manual vs. speech-based interfaces while driving four IVIS were examined. In the audio task, the driver had to select an artist, album or title. We used an MP3-player in the manual condition, whereas a laptop-based prototype music selection system was applied in the speech condition. Both systems included the same music database. Comparable selections had to be done by the drivers.

With the phone, a hands-free car kit was used which could either be operated by touch screen or via speech. The driver had to set up a call by specifying the name and type of the contact (e.g., private or work) to be called. The number of menus and the number of options for each menu was the same for the manual and the speech condition.

A first navigation system was used for two navigation tasks. In the first task, the subjects had to enter specific points-of-interests (POI) in several cities. In the second task the drivers had to enter an address. Both tasks were performed either manually on a touch screen or by voice commands. This first navigation system demanded twelve operating steps including at least six visual affirmation steps, which took approximately one minute (multiple step entry). The confirmation after every single piece of information entered (city, street name and street number separately) was carried out by visually scanning a list of potential hits and then

speaking the associated number. Thus, while this navigation system used speech input it still required some visual control for the confirmation.

Therefore, a second navigation system was included which required only six operating steps including only two visual affirmation steps. This system took only about 40 s to complete an entire address entry (single step entry). In this system, the whole address was read out loud after it has been entered and the driver had to confirm this with “yes” or “no”. Thus, this single entry navigation system used speech dialogues more fully than the first system. This system was used with the speech dialogue, only.

An observer was present during the experiment to present each of the tasks verbally. The observer noted the number of secondary tasks for each subject. However, this was mainly meant to ensure that the subjects really engaged in the secondary tasks, which all subjects did. Being qualitative data it could not reasonably be used for a statistical analysis of performance in the secondary tasks. However, the observations confirmed that the subjects engaged in the different secondary tasks in an equal manner.

### 2.3. Subjects

The subjects were 30 drivers (16 male, 14 female) who had normal or corrected to normal vision. One of these subjects was rejected because even after long training she was not able to handle the driving simulator adequately. All had a valid driver's license. The age ranged from 19 to 59 with a mean age of 33.2 (SD = 11.9). They were paid 10€ an hour for the two to three hours they took to complete the experiment.

### 2.4. Experimental design

For the experiment a within-subjects design was used. Each subject did every secondary task (audio, phone, navigation system address and point of interest), manually and via speech interface while driving the LCT which adds up to eight conditions. Additionally, as a ninth condition the second navigation system was included for address entry with the speech interface, only. The order of secondary tasks was randomized. For each secondary task (with the exception of the second navigation system which was only operated by speech), the order of the interaction mode (manual vs. speech) was also randomized.

### 2.5. Procedure

First the subjects were introduced to the experiment and signed the informed consent forms. They were then given a description of how to manage the LCT driving task. After reading the description, the experimenter explained the task again to be sure that all subjects understood the primary task. When they had completed six practice trials, subjects began the experimental scenarios. The last trial of this practice session was used as the baseline condition (driving without secondary task). Each system and task was then first introduced. The subject had to work with the system until they achieved at least two successful operations and felt comfortable handling it. Afterwards, the secondary task was combined with the LCT driving task. Each subject had to drive one trial while conducting the secondary task as a dual-task practice. Only after that, the experimental trial was driven.

The experimenter sat behind the subject and presented the tasks for the participants. For example, for the navigation system “Berlin – Rupert Street – 17” would be a task or “Singer Madonna”. Every time the subject had finished a task the next was given until one trial of the LCT was completed. After each trial the subjects answered the questions on perceived distraction described above. The whole experiment took about 2.5–3 h. After finishing

the last condition, the subjects were paid and thanked for their attendance.

## 3. Results

The aim of the study is to examine whether speech operation while driving reduces the distraction as compared to manual-visual operation. To this aim, a  $4 \times 2$  (four IVIS  $\times$  two modes of operation) repeated measures MANOVA was computed including SDLP, reaction time, mean deviation, subjective distraction and percentage gaze away from the driving task. Both main effects and the interaction were highly significant (every  $p = 0.001$ ; IVIS:  $F_{(15,246)} = 7.4$ ,  $\eta_p^2 = 0.31$ ; mode of operation:  $F_{(5,24)} = 80.5$ ,  $\eta_p^2 = 0.94$ ; interaction:  $F_{(15,246)} = 6.4$ ,  $\eta_p^2 = 0.28$ ). The effect sizes show that the main effect of the mode of operation is strongest. Thus, speech operation as compared to manual-visual operation significantly influences driver behavior and subjective evaluation. Additionally, the effects are different for the IVIS examined and, finally, the effect of speech operation depends on the IVIS analyzed.

In order to examine these effects more closely, a one-way repeated measures ANOVA was computed for each IVIS and each parameter, separately. For the three driving parameters, the baseline situation (LCT without secondary task) was included as a third condition besides speech and manual-visual operation. Thus, it was possible to analyze additionally whether speech and manual-visual operation differed from driving without a secondary task. This was especially of interest because one could assume from the multiple resource theory that speech operation might not interfere with driving as different resources are required. For the IVIS ‘navigation system address’ the second navigation system (single step entry) was added as a fourth condition (besides baseline, manual-visual and speech operation of the first navigation system). The results were corrected for sphericity violations where necessary by use of the Greenhouse–Geisser modification (Greenhouse and Geisser, 1959). Additionally, post-hoc tests (Bonferroni corrected) were done for the pair-wise comparisons. Finally,  $\eta_p^2$  is reported in order to assess the effect size. The  $\eta_p^2$  is defined as the proportion of the effect+error variance that is attributable to the effect.

For the subjective rating of distraction and the percentage of gazes away from driving the baseline situation could not be included. Distraction was not rated after driving without a secondary task. For gaze behavior, the observations yielded 0% gazes away from driving in the baseline situation. Obviously, during the three minutes of the LCT the subjects fully concentrated on the task and did not look away from the computer screen. Thus, for these two variables the one-way repeated measures ANOVAs included only two conditions (speech vs. manual-visual operation). Accordingly, post-hoc comparisons were not necessary.

As a total of 20 ANOVAs was conducted, alpha was adjusted to  $0.05/20 = 0.0025$  following the conservative procedure of Bonferroni. Despite this procedure, all but one main effect were significant indicating how strong the effect was. The results of these ANOVAs are given first as tables in order to give an overview about the structure and magnitude of the effects for each IVIS. Thus, for each IVIS it was examined whether speech operation differed from manual-visual operation and whether both differ from driving without a secondary task.

Secondly, these effects are described in figures for each parameter comparing the different IVIS. Thus, the interaction between mode of operation and IVIS can be examined more closely as well as the main effects of these factors.

Table 1 gives the results for the address entry in the navigation system. For all dependent variables, a significant main effect was found. Pair-wise post-hoc comparisons indicated that (1) speech

**Table 1**

Statistics of the main effects of the one-way repeated measures ANOVAs and the post-hoc pair-wise comparisons for the address entry. The comparison of 'manual vs. speech single' is not included as different systems were used.

| Address entry  | Main effect "modality" |          |            | Post-hoc comparisons       |                            |                     |                              |                            |
|----------------|------------------------|----------|------------|----------------------------|----------------------------|---------------------|------------------------------|----------------------------|
|                | <i>F</i>               | <i>p</i> | $\eta_p^2$ | Manual vs. speech multiple | Speech multiple vs. single | Manual vs. baseline | Speech multiple vs. baseline | Speech single vs. baseline |
| Mean deviation | 36.4                   | .001     | .565       | **                         | –                          | **                  | **                           | **                         |
| SDLP           | 12.8                   | .001     | .314       | **                         | –                          | **                  | –                            | –                          |
| Reaction time  | 33.0                   | .001     | .541       | –                          | –                          | **                  | **                           | **                         |
| %Gaze away     | 373.2                  | .001     | .930       | **                         | **                         |                     |                              |                            |
| Distraction    | 53.1                   | .001     | .655       | **                         | **                         |                     |                              |                            |

(–) Not significant. \*\*Significant alpha <0.01.

**Table 2**

Statistics of the main effects of the one-way repeated measures ANOVAs and the post-hoc pair-wise comparisons for the POI entry.

| POI entry      | Main effect "modality" |          |            | Post-hoc comparisons |                     |                     |
|----------------|------------------------|----------|------------|----------------------|---------------------|---------------------|
|                | <i>F</i>               | <i>p</i> | $\eta_p^2$ | Manual vs. speech    | Manual vs. baseline | Speech vs. baseline |
| Mean deviation | 28.3                   | .001     | .503       | –                    | **                  | **                  |
| SDLP           | 13.5                   | .001     | .326       | *                    | **                  | **                  |
| Reaction time  | 32.8                   | .001     | .539       | –                    | **                  | **                  |
| %Gaze away     | 65.8                   | .001     | .702       |                      |                     |                     |
| Distraction    | 5.9                    | .022     | .174       |                      |                     |                     |

(–) Not significant.

\* Significant alpha <0.05.

\*\* Significant alpha <0.01.

input (single and multiple steps) differed from manual control in every dependent variable except reaction time; that (2) both kinds of speech input differed from each other only in gaze behavior but not in driving behavior; and (3) manual control changed driving behavior in each dependent variable, but speech control only in mean deviation and the reaction time.

For entering a point of interest, significant main effects were found for all dependent variables (see Table 2). The post-hoc tests showed that both speech and manual control differed from baseline. However, speech control differed from manual control only for the SDLP and for the gaze parameters.

Selecting a number from an address book also led to significant main effects in all dependent variables (see Table 3). Manual control differed from baseline in all driving parameters. Speech control differed from baseline only in the mean deviation and the reaction time. Speech and manual control were different for all parameters observed.

Finally, selecting a piece of music yielded significant main effects in all dependent variables (see Table 4). At the post-hoc comparison a similar picture was found as when selecting a number. Manual control was different from speech control and baseline in every dependent variable. Speech control was comparable to baseline

**Table 3**

Statistics of the main effects of the one-way repeated measures ANOVAs and the post-hoc pair-wise comparisons for the telephone IVIS.

| Select a number | Main effect "modality" |          |            | Post-hoc comparisons |                     |                     |
|-----------------|------------------------|----------|------------|----------------------|---------------------|---------------------|
|                 | <i>F</i>               | <i>p</i> | $\eta_p^2$ | Manual vs. speech    | Manual vs. baseline | Speech vs. baseline |
| Mean deviation  | 25.3                   | .001     | .475       | **                   | **                  | **                  |
| SDLP            | 23.6                   | .001     | .457       | **                   | **                  | –                   |
| Reaction time   | 31.3                   | .001     | .528       | *                    | **                  | **                  |
| %Gaze away      | 293.0                  | .001     | .913       |                      |                     |                     |
| Distraction     | 37.2                   | .001     | .570       |                      |                     |                     |

(–) Not significant.

\* Significant alpha <0.05.

\*\* Significant alpha <0.01.

**Table 4**

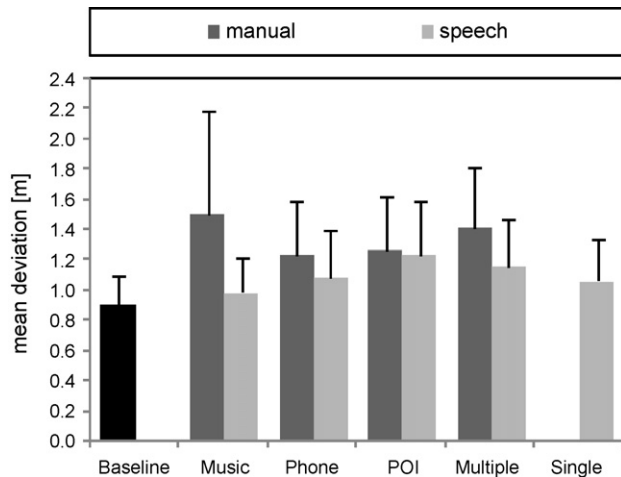
Statistics of the main effects of the one-way repeated measures ANOVAs and the post-hoc pair-wise comparisons for the music selection.

| Music selection | Main effect "modality" |          |            | Post-hoc comparisons |                     |                     |
|-----------------|------------------------|----------|------------|----------------------|---------------------|---------------------|
|                 | <i>F</i>               | <i>p</i> | $\eta_p^2$ | Manual vs. speech    | Manual vs. baseline | Speech vs. baseline |
| Mean deviation  | 27.1                   | .001     | .492       | **                   | **                  | *                   |
| SDLP            | 12.3                   | .001     | .306       | **                   | **                  | –                   |
| Reaction time   | 52.6                   | .001     | .652       | **                   | **                  | **                  |
| %Gaze away      | 291.8                  | .001     | .912       |                      |                     |                     |
| Distraction     | 328.2                  | .001     | .921       |                      |                     |                     |

(–) Not significant.

\* Significant alpha <0.05.

\*\* Significant alpha <0.01.



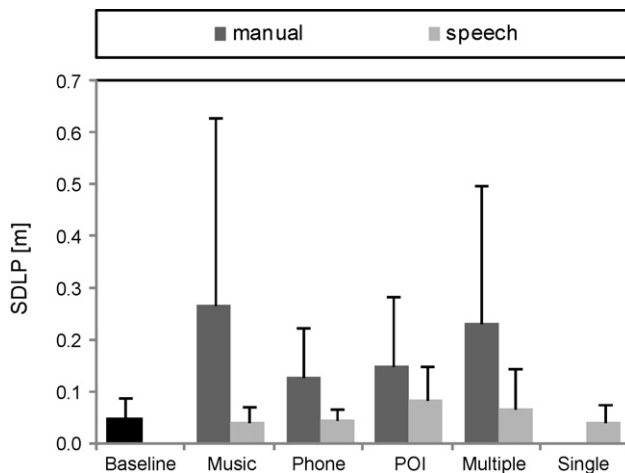
**Fig. 1.** Mean (bar) and standard deviation (whiskers) of the mean deviation for the different IVIS comparing manual vs. speech control. The baseline condition is shown as a black bar in the first column.

with regard to SDLP and differed in the mean deviation and reaction time.

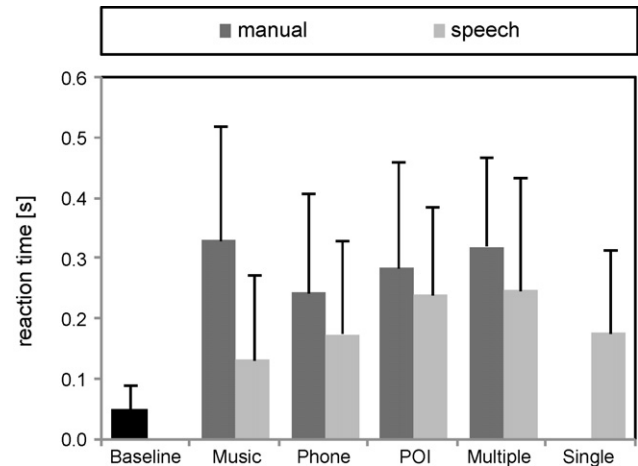
The mean deviation (see Fig. 1) as the global measure of distraction was significantly increased for every IVIS with manual control. Speech control was better than manual control for the music selection, for selecting a number and for both versions of address entry in the navigation system. For the selection of the points of interest, no advantage of the speech control could be shown. Despite this improvement in driving performance due to the speech control for the other tasks, each task with speech control was still significantly worse than the baseline condition without secondary tasks.

The picture changed a bit when looking at the lane-keeping phase of the LCT. This is the simple part of the LCT as drivers only have to remain within the lane. However, even in this easy driving-task manual control led to a significant increase in SDLP for each task (see Fig. 2). With speech control, the baseline level was accomplished with the only exception being the point-of-interest entry.

The last parameter 'reaction time' describes the driving behavior in the lane change phase of the LCT. Both manual and speech control led to a significant increase in reaction time in all tasks. An improvement due to speech control was found for music selection, selecting a number and for the single step address entry. But even in



**Fig. 2.** Mean (bar) and standard deviation (whiskers) of the SDLP for the different IVIS comparing manual vs. speech control. The baseline condition is shown as a black bar in the first column.

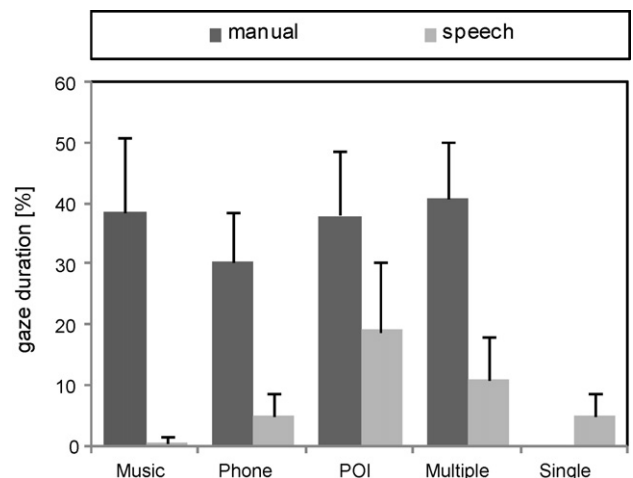


**Fig. 3.** Mean (bar) and standard deviation (whiskers) of the reaction time for the different IVIS comparing manual vs. speech control. The baseline condition is shown as a black bar in the first column.

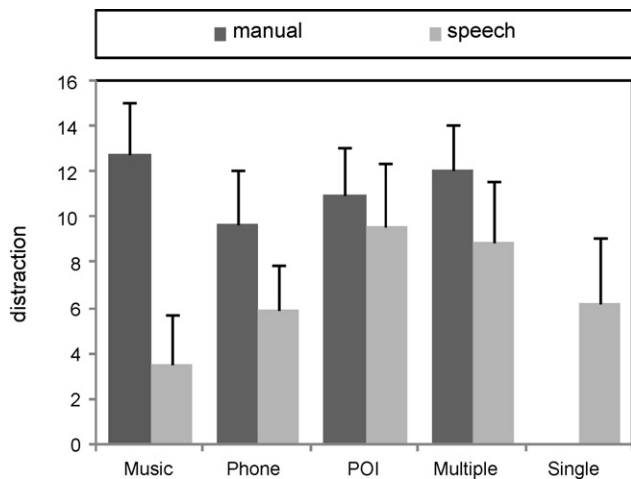
these conditions, reaction times were still longer than when driving without secondary tasks (see Fig. 3). It should be noted that reaction times were very short. This is due to the fact that drivers saw the sign in advance (but could not read it) and could then react very fast when it became legible. Moreover, it may be that they could read the sign a little bit earlier than we assumed for the analyses because this point had to be estimated. However, this analysis procedure and the estimated point were constant for all conditions.

Of course, gaze behavior played a crucial role for these effects, as the next figure shows. During manual control, 30–40% of the time was used to look at the displays (see Fig. 4). Depending on the tasks, the mean duration was between 0.7 and 1 s. The percentage of gaze time away from LCT screen was substantially reduced by the speech control. This percentage was largest for the speech entry of points of interest, followed by the multiple step procedure of address entry.

Finally, as Fig. 5 shows, subjects were well aware that the different tasks distracted them. The subjective distraction was stronger for manual control than for speech control. But subjects also knew that even the least distracting task using speech selection (music selection) still presented some low level of distraction (which could also be demonstrated in the driving performance).



**Fig. 4.** Mean (bar) and standard deviation (whiskers) of the percentage of gaze time away from the LCT screen for the different IVIS comparing manual vs. speech control. The baseline condition is not shown as it was 0%.



**Fig. 5.** Mean (bar) and standard deviation (whiskers) of the subjective distraction for the different IVIS comparing manual vs. speech control. This question was obviously not answered in the baseline condition.

#### 4. Discussion

This study examined whether speech interfaces for different IVIS systems used in the car reduce the distraction that these systems are known to produce. Several typical systems (audio, phone number selection, navigation system with address entry and point-of-interest selection) were compared with speech and manual control, each. In order to measure driving performance, the Lane Change Task was used. Additionally, gaze behavior and a subjective measure of distraction were examined.

First of all, this experimental setting was very well able to demonstrate the strong distraction effect caused by the manual control of the different systems. Lane-keeping quality was substantially reduced and the reaction time needed to change the lane was increased when drivers manually controlled different systems. This was mainly due to visual distraction as 30–40% off-the-screen glance time show. The results are very similar to those of Chisholm *et al.* (2008) regarding the MP3 player and of Reed and Robbins (2008) regarding the phone condition.

Speech interfaces improved driving performance for all systems with the exception of point-of-interest entry in the navigation system. This may be due to the fact that this point-of-interest entry required multiple visual confirmations and thus had the largest percentage of gazes away from the LCT screen. For the other systems, speech control effectively decreased gazes away from the driving task. Thus, speech enables the drivers to focus more on the road and this improves driving performance.

However, these improvements were not strong enough to reach the baseline performance level in all parts of the driving task. Even with speech control, all systems lead to a larger mean deviation from the ideal trajectory of the car in the LCT. These effects were especially strong in the lane-change phase of the task. During lane-keeping, the easier of the speech-controlled tasks did not change driving performance significantly. But the effects were clearly visible even for these tasks during the lane change phase leading to increased reaction times. Moreover, while subjects found the tasks with speech control less distracting than with manual control, they still indicated some low level of distraction even with speech control.

Thus, on the one hand speech control clearly reduces the visual distraction introduced by manual control of different systems typically used in the car. Drivers are aware that these systems distract them but still keep on using them. One could hope that drivers engage in using these systems only if the driving situation is easy

and nothing is likely to happen. As it was shown that the distracting effect is minimal when the driving situation is easy, this could result in a safe handling as long as the drivers remain sensible. However, even then the traffic situation may change very fast. Increased reaction times are then to be expected even with speech control. The data of Klauer *et al.* (2006) can be interpreted in this way. They estimated that these secondary tasks substantially increased the risk of a critical situation or an accident.

The design of the verbal human–machine-interaction seems to be relevant with regard to distraction. The two navigation systems with different concepts for the speech interface showed clearly different effects with a single entry and verbal feedback having advantages with regard to driving performance. But it may also be that the secondary task itself is more or less cognitive demanding for the driver. If the task is becoming more and more complex, even a simple to use speech interface cannot compensate for that. Thus, it will be crucial for future systems for in-car use to keep the cognitive complexity at an adequate level which still can be handled by the driver. But when looking at the strong negative effects of manual control and the benefits of the speech interfaces examined, speech control is a must in the car of the future.

Finally, as this study was conducted in a laboratory, drivers were not exposed to the same risk as on an actual road. Awareness of the lack of severe consequences may affect how drivers prioritize their attention. It is reasonable to assume that drivers might be more cautious when really driving. On the other hand, the LCT is a quite simple driving task. In real driving, task difficulty may be much larger, e.g. at complex intersections, when negotiating the right of way, when merging on highways with dense traffic etc. Thus, the effects of secondary tasks may be underestimated when using the LCT. From these reasons it is hard to predict how much certain secondary tasks will influence accident rates. However, for the basic mechanisms leading to distraction and the comparison of different tasks the LCT does probably give a good indication.

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